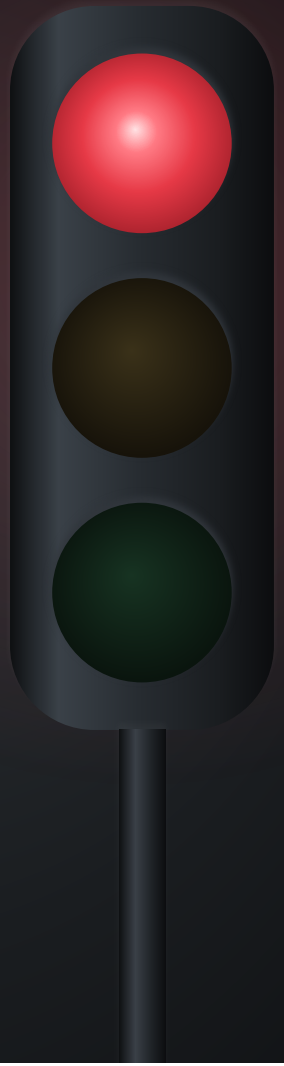
UNFALLATLAS 2025 · GERMANY

How likely is an injury accident to be severe?

Predicting severity before it happens, for prevention planning



THE RANDOM FOREST RANGERS

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Planners need to know where accidents are most likely to end badly

PROBLEM

A regional road-safety authority decides **where and when** to invest in prevention.

TYPICAL PREVENTION TOOLS (EXAMPLES)

01

Speed limits

02

Safer curves and crossings

03

Targeted campaigns

Our data: every police-recorded injury accident in Germany, 2025 (Unfallatlas)

We estimate severity before an accident, so how it happened stays out

PROBLEM

OUR QUESTION

If an injury accident happens here, how likely is it to be severe?

Severe = someone killed or seriously injured. We do not predict whether accidents happen.

KNOWN BEFOREHAND · USED BY OUR MODEL

- Place
- Time
- Light
- Road surface
- Road users

KNOWN ONLY AFTERWARDS · NOT USED

- **Accident kind and type**
  e.g. “head-on collision”

IN SHORT **Our model finds 58% of severe accidents on unseen data.**

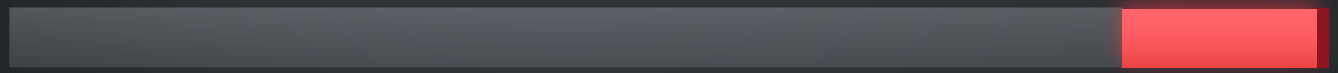
Test set, 40,927 accidents, used once.

About 1 in 6 injury accidents is severe, so simple accuracy would mislead

PROBLEM

272,812

police-recorded injury accidents, 2025 (195 without population data removed)



84.3% slightly injured

14.8% seriously injured · **0.9%** killed

15.7%

SEVERE

ADDED

Municipality statistics: population density, city / town / rural

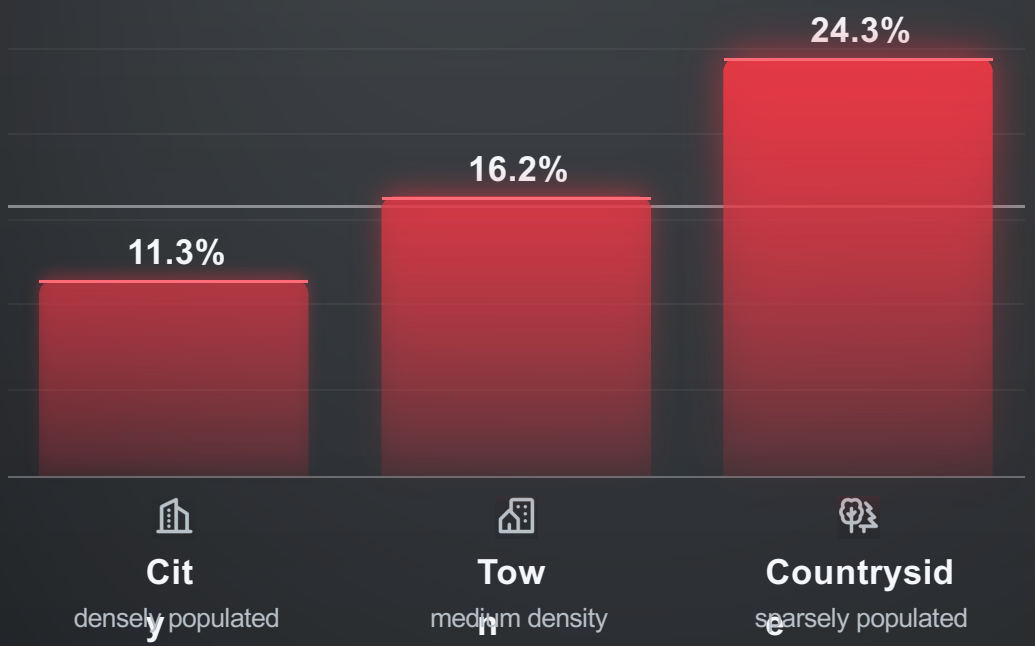
SPLIT

70% train | 15% | 15%

Validation · test (used once, at the end)

Severe accidents are twice as common in the countryside as in cities

FINDINGS



Severe share by municipality type. Dashed line: average 15.7%. All 2025 accidents, 272,812.

Of the 13-point urban–rural gap, about 10.5 points come from the same kind of accident being more severe in rural areas.

Likely causes: higher speeds, longer rescue times. Gap measured on all 2025 accidents.

Motorcycles, nights and weekends carry the highest risk of a severe outcome

FINDINGS

SEVERE SHARE WHEN INVOLVED



35.1% Motorcycle in a rural municipality

SEVERE SHARE BY HOUR OF DAY



Saturday	Sunday	Mon–Fri
17.9%	19.8%	14.2–15.6%

Training data, 190,970 accidents. Dashed line: average.

A model is only useful if it beats two simple rules of thumb

MODEL

TWO PLAIN QUESTIONS

How many severe accidents does it find?
Recall

How many of its alarms are really severe?
Precision

F1 One score that balances both, from 0 to 1. Higher is better.

BASELINES TO BEAT

Rule
Motorcycle or pedestrian involved, or no car

F1
0.303

Historical lookup
Severe share of similar past accidents

F1
0.315

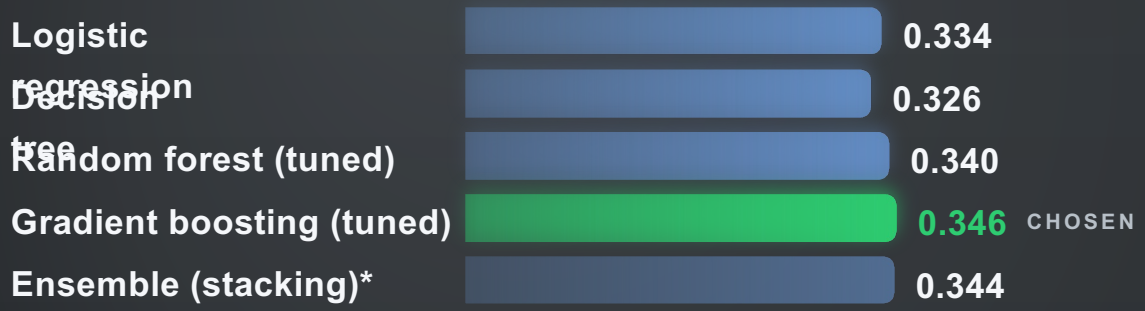
Validation set, 40,915 accidents.

On the test set: 0.300 and 0.312. Test set, 40,927 accidents, used once.

Every approach we tried ends at about the same score

MODEL

F1 AFTER TUNING, SCALE 0 TO 0.4



Chosen by the best cross-validated score on training data (PR-AUC 0.267), not by validation, so the choice isn't fitted to one data set.

The limit is the information in the data, not the algorithm.

Validation set, 40,915 accidents. Tuning changed almost nothing (≤ 0.001). *Stacking: from the team's other final notebook (Micheal), same data and split; combining models added at most +0.003.

WHAT MATTERS MOST

Who is involved

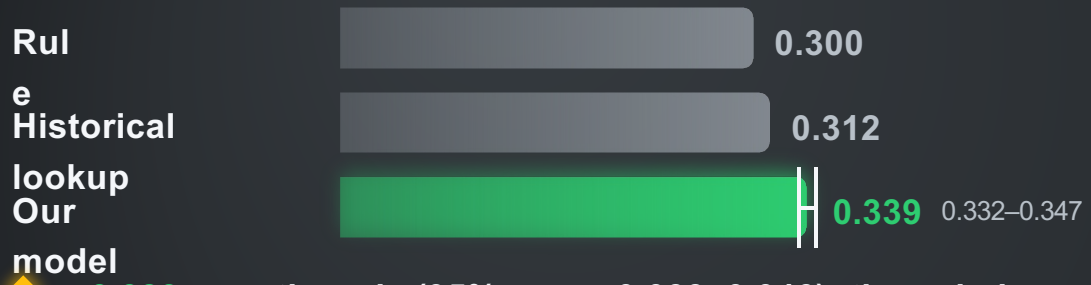
How rural the place is

Population density is by far the most important factor (about 4–5× more than the next), followed by motorcycle and car involvement.

Our model beats both baselines, but the gain is modest

MODEL

F1 SCORE · TEST SET, USED ONCE



+0.039 over the rule (95% range 0.033–0.046): the gain is real, not luck

+0.027 over the historical lookup

HOLDS UP

Validation 0.346 → Test 0.339: stable on unseen data

Test set, 40,927 accidents, used once after all choices were fixed. 95% ranges from 1,000 bootstrap resamples. PR-AUC 0.268.

58 of 100 — **severe accidents found**
Test set

1 in 4 — **flagged accidents really severe**
Test set · average 1 in 6

1 in 3 — **of the 10% highest-risk accidents are severe (33%), about twice the average**
Historical lookup: 27% · Validation set, 40,915 accidents

−17% — **false alarms vs the historical lookup, finding almost as many severe accidents**
Validation set, 40,915 accidents

The predicted percentages match what really happens

MODEL

15.6%

PREDICTED

≈

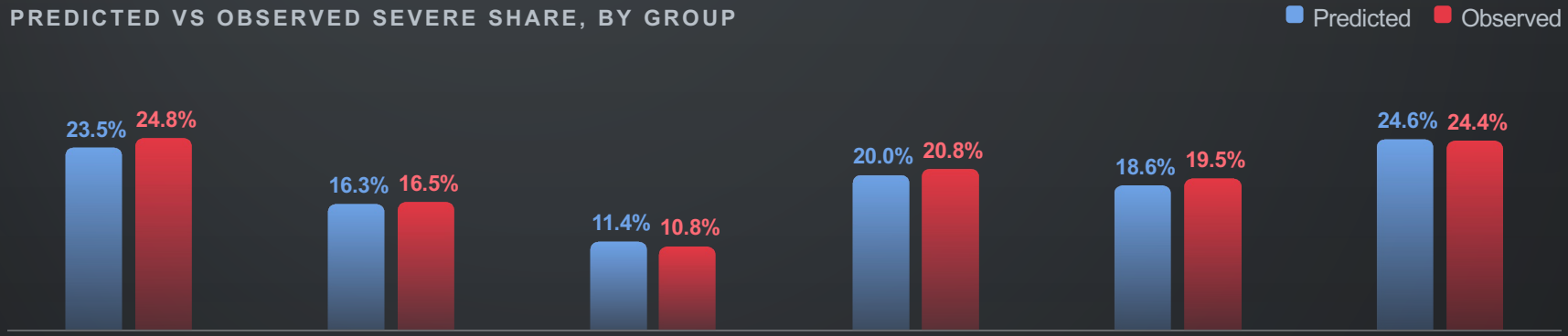
15.7%

OBSERVED

All groups within about 1.3 percentage points.

Trained with normal class weights, so no correction step was needed.

PREDICTED VS OBSERVED SEVERE SHARE, BY GROUP



Validation set, 40,915 accidents.

The model finds most severe accidents in the countryside but few in cities

MODEL



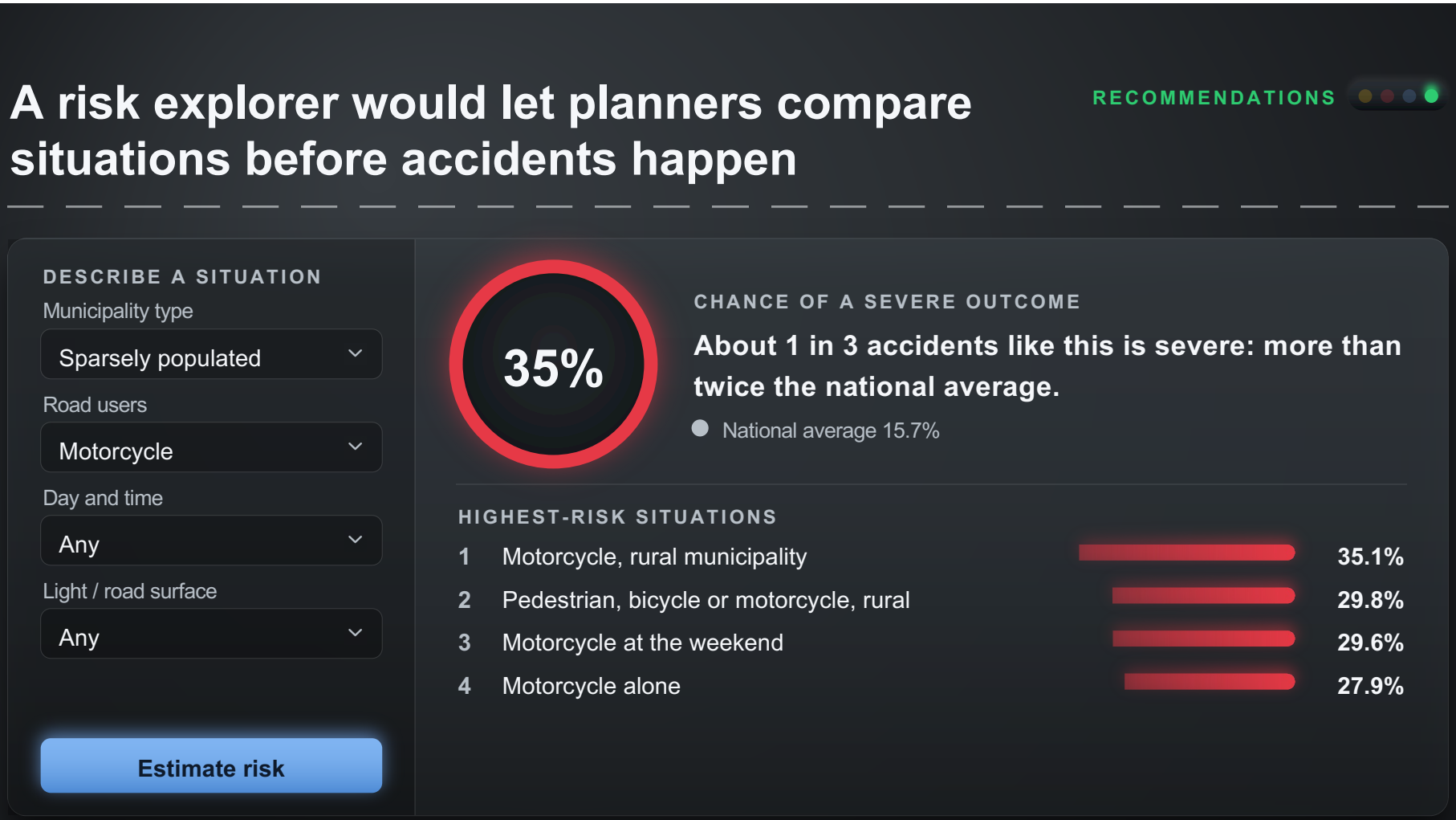
Validation set, 40,915 accidents, one national threshold (0.167).

Risks per group are still correct. City accidents have a lower risk (about 11%), so they rarely pass one national threshold: about **2 in 3** severe city accidents are missed.

- **Strong for rural prevention. Cities need their own threshold or model.**

A risk explorer would let planners compare situations before accidents happen

RECOMMENDATIONS



Illustrative mock-up; values are observed severe shares, 2025 training data. For prioritising prevention, not for automatic decisions.

Prevention should start on rural roads, especially for motorcycles

RECOMMENDATIONS

01

Prioritise rural roads, especially for motorcycles

Here more than 1 in 3 motorcycle accidents is severe, so prevention reaches the most serious cases.

02

Use the model as a risk ranking

Read the percentages; don’t treat it as a yes/no decision

03

Give cities their own threshold or model

One national threshold finds only about 1 in 3 severe city accidents.

Our data shows where risk is highest, not which measure works best.

Real road and speed data is the most promising next step

RECOMMENDATIONS

LIMITS

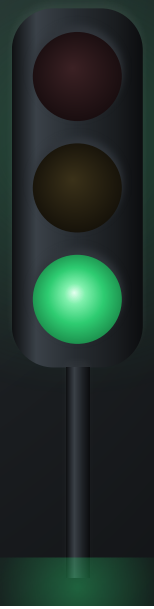
- No speed, road type, helmet, seatbelt or vehicle data
- One year only (2025)
- Density describes the municipality, not the road
- Patterns, not causes

NEXT STEPS

- Road type and speed limits (e.g. OpenStreetMap)
- Separate urban and rural thresholds
- Several years of data

Thank you. Questions?

THE RANDOM FOREST RANGERS



The model uses 20 features, all known before an accident

BACKUP

Road users — 9

Six flags (bicycle, car, pedestrian, motorcycle, truck, other vehicle), vulnerable road user involved, vulnerable vs vehicle, number of road-user types

Place — 4

State, latitude, longitude, population density (log)

Time — 5

Hour and month (cyclic), weekend

Conditions — 2

Light, road surface

Not used: accident kind and type (known only afterwards)

Official urbanisation (city / town / rural) is used to explain results, not in the model (see B3).

Every choice was fixed before the test set was used

BACKUP

01

Split

70% training · 15% validation · 15% test

190,970 · 40,915 · 40,927 accidents

02

Choose the model

Chosen by cross-validated PR-AUC on the training data (0.267)

Training data, 190,970 accidents

03

Set the threshold

0.167, chosen on out-of-fold predictions of the training data

Training data, 190,970 accidents

04

Compare

Against the rule and the historical lookup: F1 0.346

Validation set, 40,915 accidents

Test, used once: F1 0.339 (0.332–0.347), PR-AUC 0.268, vs 0.346 on validation

Test set, 40,927 accidents, used once. 95% range from 1,000 bootstrap resamples.

Which city-vs-countryside measure should the model use?

BACKUP

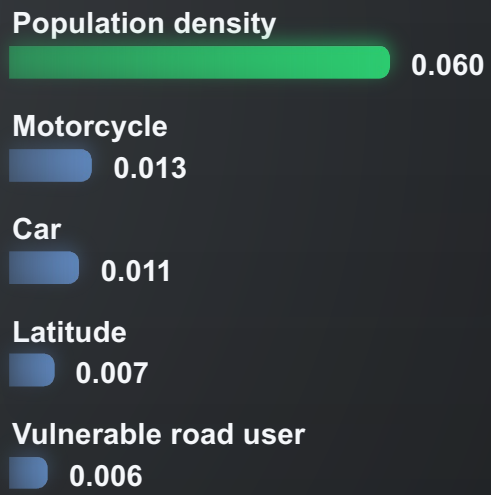
CROSS-VALIDATED PR-AUC, SCALE 0 TO 0.3



City-vs-countryside information is a clear gain. Density alone is as good as both together, so we kept the simpler model. Urbanisation and density are strongly related (r = −0.88).

Training data, 190,970 accidents. Noise ≈ ±0.004.

MOST IMPORTANT FEATURES



Drop in PR-AUC when shuffled. Validation set, 40,915 accidents.

Predicting before the accident costs only a little accuracy

BACKUP

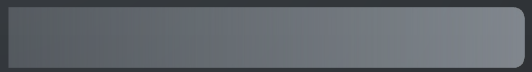
F1 SCORE · VALIDATION

Our final model
known beforehand only



0.346

Reference model also uses accident kind and type



0.375

Validation set, 40,915 accidents. Reference from a comparable gradient boosting model in the team's final notebook (Micheal), validation set.

◆ **In the reference model, leaving out how the accident happened costs about 0.029 : the price of being usable for planning.**